PART II - VERSION 2 INDUSTRIAL PLC PROJECTS

Project 2 - Industrial Batch Conveyor Counter System

Version 2 - Industrial PLC Upgrade

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

Project 2 Version 2 upgrades the original conveyor batch-counting project into a more structured PLC-controlled conveyor system with Auto/Manual operation, operating permissives, startup delay, configurable batch quantity, batch-complete control, jam monitoring, fault handling, and operator status indication.

I developed the project using an Allen-Bradley MicroLogix 1100 PLC and RSLogix Micro Starter Lite. The final program separates the operator request to run the conveyor from the physical Conveyor_Motor O:2/0 output. Auto and Manual requests are processed through System_Ready and common Conveyor_Run logic before a three-second startup delay permits the conveyor motor to energize.

Before operation is allowed, the PLC monitors EStop_OK I:1/5 and Overload_OK I:1/6, the master Fault_Active B3:0/4 state, and the Auto/Manual mode selection. Simultaneous selection of Auto_Mode I:1/3 and Manual_Mode I:1/4 creates Mode_Conflict B3:0/5, which participates in the master fault architecture.

In Auto mode, a valid Start_PB I:1/0 command latches Auto_Run_Request B3:0/0. Manual mode uses the non-latched Manual_Run_Request B3:0/1. The two requests are combined into Conveyor_Run B3:0/2 so both modes share the same startup-delay and motor-control logic.

The production-counting section uses Target Quantity N7:0 as the configurable batch target. An unconditional MOV transfers N7:0 into C5:0.PRE. While Conveyor_Motor is operating, Photoeye I:1/7 and ONS B3:2/0 allow C5:0 to increment once per detected product. When the counter reaches its preset, Batch_Complete B3:0/6 latches, stops the conveyor through the motor-control logic, and energizes Batch_Complete_Lamp O:2/4.

Jam monitoring is implemented with T4:1. While the conveyor is running and Photoeye remains false, the timer accumulates toward a ten-second preset. If that interval expires, Jam_Fault B3:1/0 latches and becomes a source for the master Fault_Active condition.

The project was developed and tested as a PLC portfolio/training implementation. The PLC monitoring of E-stop and overload status represents standard control-system monitoring and does not replace safety-rated hardware or constitute machine-safety validation.

2. Engineering Objective

The objective of Project 2 Version 2 was to redesign the original conveyor counter so that product counting became part of a complete conveyor-control sequence rather than an isolated sensor-and-counter exercise. I retained the core batch-counting function while adding operating modes, request management, startup timing, batch states, diagnostics, and controlled recovery.

- Provide separate Auto and Manual operating modes.
- Detect simultaneous Auto/Manual selection.
- Establish centralized Start_Permissive and System_Ready conditions.
- Separate operator run requests from the physical Conveyor_Motor output.
- Provide maintained automatic operation and non-latched manual operation.
- Apply a three-second startup delay before conveyor energization.
- Prevent operation after completion of the required production batch.
- Allow the batch quantity to be configured through N7:0.
- Count valid Photoeye events only while the conveyor is operating.
- Use a one-shot so one sustained sensor event produces one count.
- Latch Batch_Complete when the configured quantity is reached.

- Provide a dedicated Batch_Complete_Lamp.
- Detect extended conveyor operation without product detection.
- Latch Jam_Fault after the configured timeout.
- Combine the implemented abnormal conditions into a master Fault_Active state.
- Provide deliberate reset/recovery and an audible master-fault indication.

3. Functional Requirements

- The PLC shall detect simultaneous Auto_Mode I:1/3 and Manual_Mode I:1/4 selection and energize Mode_Conflict B3:0/5.
- Mode_Conflict, an unhealthy EStop_OK I:1/5 input, an unhealthy Overload_OK I:1/6 input, or Jam_Fault B3:1/0 shall be able to latch Fault_Active B3:0/4.
- Reset_PB I:1/2, together with healthy E-stop and overload conditions and no Mode_Conflict, shall unlatch Fault_Active and Jam_Fault through the implemented reset rung.
- Start_Permissive B3:0/7 shall require Fault_Active to be false, EStop_OK and Overload_OK to be true, and Mode_Conflict to be false.
- System_Ready B3:0/3 shall require Start_Permissive together with either Auto_Mode or Manual_Mode.
- In Auto mode, Start_PB and System_Ready shall latch Auto_Run_Request B3:0/0 when Batch_Complete is false.
- Auto_Run_Request shall unlatch if Stop_OK I:1/1 becomes false, Fault_Active becomes true, Auto_Mode is removed, or Batch_Complete becomes true.
- In Manual mode, Start_PB and System_Ready shall energize Manual_Run_Request B3:0/1 only while Batch_Complete is false and all enabling conditions remain true.
- Auto_Run_Request or Manual_Run_Request, together with System_Ready, shall energize Conveyor_Run B3:0/2.
- Conveyor_Run shall enable Startup_Delay T4:0 with time base 1.0 second and preset 3.
- Conveyor_Motor O:2/0 shall energize only when T4:0/DN is true, Conveyor_Run remains true, and Batch_Complete is false.
- Running_Lamp O:2/1 shall follow Conveyor_Motor.
- An unconditional MOV shall transfer N7:0 into C5:0.PRE.
- While Conveyor_Motor is true, a valid Photoeye I:1/7 transition shall pass through ONS B3:2/0 and increment C5:0 once.
- C5:0/DN shall latch Batch_Complete B3:0/6.
- Batch_Complete shall energize Batch_Complete_Lamp O:2/4 and inhibit further conveyor operation until reset.
- Reset_PB together with Batch_Complete shall reset C5:0 and unlatch Batch_Complete.
- While Conveyor_Motor is true and Photoeye is false, Jam_Timer T4:1 shall accumulate with time base 1.0 second and preset 10.
- T4:1/DN shall latch Jam_Fault B3:1/0.
- Fault_Active shall energize Alarm_Horn O:2/3.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes conveyor control, counting, fault, and monitoring logic
Programming Software	RSLogix Micro Starter Lite 8.30	PLC programming, simulation, and online monitoring
Start Push Button	I:1/0	Provides operator start command
Stop Healthy Input	I:1/1	Removes the maintained Auto run request when the healthy signal is lost
Reset Push Button	I:1/2	Provides fault and batch reset command
Auto Mode Selector	I:1/3	Selects automatic operation
Manual Mode Selector	I:1/4	Selects manual operation
E-stop Healthy Input	I:1/5	Provides E-stop circuit healthy status
Overload Healthy Input	I:1/6	Provides overload healthy status
Photoeye	I:1/7	Detects products for counting and jam monitoring
Conveyor Motor	O:2/0	Commands conveyor operation
Running Lamp	O:2/1	Indicates commanded conveyor operation
Alarm Horn	O:2/3	Provides audible master-fault indication
Batch Complete Lamp	O:2/4	Indicates completed production batch
Startup Delay	T4:0	Provides three-second startup delay
Jam Timer	T4:1	Detects prolonged operation without Photoeye detection
Batch Counter	C5:0	Accumulates product counts
Target Quantity	N7:0	Provides configurable batch-counter preset

The final ladder does not contain a separate Fault Lamp output rung. Although an earlier supporting document listed one, it is not included in this final implementation and is therefore not claimed here.

5. I/O Assignment

Physical Inputs

Device	PLC Address	Type	Description
Start_PB	I:1/0	Digital Input	Operator start command
Stop_OK	I:1/1	Digital Input	Stop circuit healthy signal used by Auto request reset
Reset_PB	I:1/2	Digital Input	Fault/batch reset command
Auto_Mode	I:1/3	Digital Input	Auto-mode selector
Manual_Mode	I:1/4	Digital Input	Manual-mode selector
EStop_OK	I:1/5	Digital Input	E-stop circuit healthy
Overload_OK	I:1/6	Digital Input	Overload healthy

Photoeye	I:1/7	Digital Input	Product-detection input

Physical Outputs

Device	PLC Address	Type	Description
Conveyor_Motor	O:2/0	Digital Output	Conveyor motor command
Running_Lamp	O:2/1	Digital Output	Conveyor-running indication
Alarm_Horn	O:2/3	Digital Output	Audible master-fault indication
Batch_Complete_Lamp	O:2/4	Digital Output	Batch-completion indication

Internal PLC Data

Function	Address	Type	Purpose
Auto_Run_Request	B3:0/0	Internal Bit	Latched automatic operating request
Manual_Run_Request	B3:0/1	Internal Bit	Non-latched manual operating request
Conveyor_Run	B3:0/2	Internal Bit	Common conveyor run request
System_Ready	B3:0/3	Internal Bit	System ready status
Fault_Active	B3:0/4	Latched Internal Bit	Master fault state
Mode_Conflict	B3:0/5	Internal Bit	Simultaneous Auto/Manual selection
Batch_Complete	B3:0/6	Latched Internal Bit	Completed batch state
Start_Permissive	B3:0/7	Internal Bit	Master startup permission
Jam_Fault	B3:1/0	Latched Internal Bit	Jam diagnostic fault
Photoeye One-Shot	B3:2/0	ONS Storage Bit	One product count per sensor transition
Startup_Delay	T4:0	TON	Three-second startup delay
Jam_Timer	T4:1	TON	Ten-second no-product timer
Batch_Counter	C5:0	CTU	Accumulated product count
Target Quantity	N7:0	Integer	Configurable batch target

6. Control Strategy

I organized the Version 2 conveyor program around operating validity, run-request management, delayed startup, batch production control, and jam/fault monitoring.

The first part of the ladder determines whether operation is valid. Simultaneous Auto and Manual selection creates Mode_Conflict. Fault_Active latches from Mode_Conflict, loss of the E-stop healthy signal, loss of the overload healthy signal, or Jam_Fault. Start_Permissive then provides one common status representing the implemented healthy/fault conditions.

System_Ready requires Start_Permissive plus an Auto or Manual mode selection. Auto operation uses a latched Auto_Run_Request; its dedicated unlatch rung responds to Stop_OK loss, Fault_Active, Auto-mode removal, or Batch_Complete. Manual operation uses a non-latched Manual_Run_Request. In the final ladder, Stop_OK is specifically part of the Auto request reset path; it is not placed in Start_Permissive or the Manual request rung, so I document its function exactly as implemented rather than treating it as a universal permissive.

The mode-specific requests are combined into Conveyor_Run. This common signal starts T4:0. After three seconds, Conveyor_Motor can energize if Conveyor_Run remains true and Batch_Complete remains false.

The production-counting section separates the target from the CTU instruction. An unconditional MOV continuously copies N7:0 into C5:0.PRE. Products are counted only while Conveyor_Motor is ON. Photoeye is processed through ONS B3:2/0 so one sustained product detection does not create repeated counts.

When C5:0 reaches its preset, Batch_Complete latches. That state inhibits the motor, prevents a new Auto request, removes an existing Auto request, inhibits the Manual request, and energizes the Batch Complete Lamp. Reset first clears C5:0 and then unlatches Batch_Complete in the following rung.

Jam monitoring uses conveyor state and sensor inactivity. While the conveyor is running and Photoeye remains false, T4:1 accumulates. If the ten-second interval expires, Jam_Fault latches. Jam_Fault then latches Fault_Active, which removes Start_Permissive and produces the common alarm response.

7. Engineering Design Decisions

Design Decision	Engineering Reason
Separate Auto and Manual modes	Supports maintained automatic operation and non-latched manual operation
Dedicated Mode_Conflict bit	Makes simultaneous mode selection an explicit abnormal condition
Latched Fault_Active	Combines major abnormal conditions into one persistent master fault
Central Start_Permissive	Provides one diagnostic point for the implemented healthy/fault conditions
Separate System_Ready	Distinguishes general permission from readiness with an operating mode selected
Latched Auto_Run_Request	Allows a momentary Start command to establish maintained automatic operation
Dedicated Auto request reset rung	Makes Stop, fault, mode removal, and Batch Complete shutdown conditions explicit
Non-latched Manual_Run_Request	Requires the manual command conditions to remain present
Common Conveyor_Run bit	Allows Auto and Manual modes to share startup and motor logic
Three-second Startup_Delay	Prevents immediate conveyor energization following a valid request
Batch_Complete inhibits motor	Stops production at the configured target quantity
N7:0 moved to C5:0.PRE	Makes the batch quantity configurable without editing the CTU rung
Motor-qualified Photoeye counting	Prevents counts while the conveyor command is OFF
ONS before C5:0	Generates one count per Photoeye transition
Latched Batch_Complete	Preserves the end-of-batch state until deliberate reset
Separate C5:0 reset and Batch_Complete unlatch	Makes the reset sequence and scan order easy to trace

Ten-second Jam_Timer	Detects extended running without product detection
Latched Jam_Fault	Preserves the diagnostic event until reset
Batch_Complete_Lamp	Provides direct operator indication of completed production
Alarm_Horn from Fault_Active	Provides one audible response for the implemented master faults

A meaningful Version 2 improvement is that Batch_Complete is an operating state rather than only an indication. It participates in the Auto request, Auto reset, Manual request, and motor-output logic, so the completed batch is integrated throughout the control sequence.

Another important improvement is the MOV from N7:0 to C5:0.PRE. This separates the production target from fixed ladder configuration and creates a clear future path for HMI quantity entry.

8. Ladder Logic Description

The final Project 2 Version 2 program contains 21 functional rungs, Rung 0000 through Rung 0020, followed by the END rung. Each typed representation below is reconstructed from the final RSLogix screenshots. The final screenshot controls the addresses, instructions, branches, timer settings, counter/register usage, and symbolic names.

8.1 Rung 0000 - Mode Conflict Detection

Ladder Logic:

```
|----[ XIC I:1/3 ]----[ XIC I:1/4 ]--------------------( OTE B3:0/5 )----|
       Auto_Mode          Manual_Mode                          Mode_Conflict
```

Rung Description: When Auto_Mode I:1/3 and Manual_Mode I:1/4 are both true, Mode_Conflict B3:0/5 energizes. Because the output is an OTE, the bit follows the current simultaneous-selection condition and clears when the conflict disappears.

Engineering Purpose: I included this rung so simultaneous Auto/Manual selection is explicitly identified and can be used by both the master-fault and start-permissive logic.

8.2 Rung 0001 - Master Fault Latch

Ladder Logic:

```
     +----[ XIC B3:0/5 ]-------------------------------+
     |     Mode_Conflict                               |
     +----[ XIO I:1/5 ]--------------------------------+
     |     EStop_OK                                    |
|----+----[ XIO I:1/6 ]--------------------------------+----( OTL B3:0/4 )----|
     |     Overload_OK                                 |     Fault_Active
     +----[ XIC B3:1/0 ]-------------------------------+
           Jam_Fault
```

Rung Description: Any Mode Conflict, unhealthy E-stop status, unhealthy overload status, or active Jam Fault makes one branch true and latches Fault_Active B3:0/4. OTL preserves the master fault until the reset rung executes.

Engineering Purpose: I used one latched master fault so the rest of the program can respond consistently to several abnormal conditions.

8.3 Rung 0002 - Fault and Jam Reset

Ladder Logic:

```
|----[ XIC I:1/2 ]----[ XIC I:1/5 ]----[ XIC I:1/6 ]----[ XIO B3:0/5 ]----+----( OTU B3:0/4 )----|
      Reset_PB          EStop_OK         Overload_OK      Mode_Conflictnnn |       Fault_Active
                                                                           |
                                                                           +----( OTU B3:1/0 )----|
                                                                                  Jam_Fault
```

Rung Description: Reset is accepted only while Reset_PB is true, EStop_OK and Overload_OK are healthy, and Mode_Conflict is false. The two output branches unlatch Fault_Active and Jam_Fault during the same scan.

Engineering Purpose: I conditioned reset on restored healthy conditions so the operator cannot clear the stored fault state while the monitored abnormal condition is still present.

8.4 Rung 0003 - Start Permissive

Ladder Logic:

```
|----[ XIO B3:0/4 ]----[ XIC I:1/5 ]----[ XIC I:1/6 ]----[ XIO B3:0/5 ]----( OTE B3:0/7 )----|
     Fault_Active         EStop_OK         Overload_OK      Mode_Conflict      Start_Permissive
```

Rung Description: Start_Permissive B3:0/7 is true only when Fault_Active is false, EStop_OK and Overload_OK are true, and Mode_Conflict is false. Stop_OK is not present in this rung in the final ladder.

Engineering Purpose: I centralized the implemented healthy/fault conditions in one permissive bit so readiness and start logic can reference a clear diagnostic status.

8.5 Rung 0004 - System Ready

Ladder Logic:

```
                                    +----[ XIC I:1/3 ]----+
                                    |     Auto_Mode       |
|----[ XIC B3:0/7 ]-----------------+                     +----( OTE B3:0/3 )----|
       Start_Permissive             |                     |      System_Ready
                                    +----[ XIC I:1/4 ]----+
                                          Manual_Mode
```

Rung Description: System_Ready B3:0/3 requires Start_Permissive plus either Auto_Mode or Manual_Mode. Simultaneous mode selection is already handled by Mode_Conflict and Start_Permissive.

Engineering Purpose: This separates general operating permission from the status indicating that a mode has actually been selected.

8.6 Rung 0005 - Auto Run Request Latch

Ladder Logic:

```
|----[ XIC I:1/3 ]----[ XIC I:1/0 ]----[ XIC B3:0/3 ]----[ XIO B3:0/6 ]----( OTL B3:0/0 )----|
      Auto_Mode          Start_PB          System_Ready       Batch_Complete
Auto_Run_Request
```

Rung Description: In Auto mode, pressing Start while System_Ready is true and Batch_Complete is false latches Auto_Run_Request B3:0/0. The request remains true after Start_PB is released.

Engineering Purpose: I used a latched internal Auto request so the momentary Start command establishes maintained operation without directly latching the motor output.

8.7 Rung 0006 - Auto Run Request Reset

Ladder Logic:

```
     +----[ XIO I:1/1 ]--------------------------------+
     |     Stop_OK                                     |
     +----[ XIC B3:0/4 ]-------------------------------+
     |     Fault_Active                                |
|----+----[ XIO I:1/3 ]--------------------------------+----( OTU B3:0/0 )----|
     |     Auto_Mode                                   |     Auto_Run_Request
     +----[ XIC B3:0/6 ]-------------------------------+
           Batch_Complete
```

Rung Description: Any branch can unlatch Auto_Run_Request: Stop_OK becomes false, Fault_Active becomes true, Auto_Mode is removed, or Batch_Complete becomes true.

Engineering Purpose: This rung clearly defines the shutdown conditions for a maintained Auto request and prevents the stored request from surviving those events.

8.8 Rung 0007 - Manual Run Request

Ladder Logic:

```
|----[ XIC I:1/4 ]----[ XIC I:1/0 ]----[ XIC B3:0/3 ]----[ XIO B3:0/6 ]----( OTE B3:0/1 )----|
      Manual_Mode        Start_PB         System_Ready      Batch_Complete      Manual_Run_Request
```

Rung Description: Manual_Run_Request B3:0/1 remains true only while Manual_Mode, Start_PB, System_Ready, and the not-Batch-Complete condition remain true. The rung uses OTE rather than OTL.

Engineering Purpose: I used non-latched Manual operation so the manual request is removed as soon as its enabling command conditions are removed.

8.9 Rung 0008 - Common Conveyor Run Request

Ladder Logic:

```
     +----[ XIC B3:0/0 ]----+
     |     Auto_Run_Request |
|----+                      +-------------[ XIC B3:0/3 ]------( OTE B3:0/2 )----|
     |                      |                System_Ready       Conveyor_Run
     +----[ XIC B3:0/1 ]----+
           Manual_Run_Request
```

Rung Description: Either Auto_Run_Request or Manual_Run_Request can establish Conveyor_Run B3:0/2, but System_Ready must also remain true.

Engineering Purpose: I combined the mode-specific requests before the common timing and motor logic to avoid duplicating downstream control functions.

8.10 Rung 0009 - Startup Delay

Ladder Logic:

```
|----[ XIC B3:0/2 ]------------------------------------[ TON T4:0 ]----|
      Conveyor_Run                                      Startup_Delay   Time Base: 1.0  PRE: 3
```

Rung Description: When Conveyor_Run remains true, Startup_Delay T4:0 accumulates. After three seconds T4:0/DN becomes true. Loss of Conveyor_Run resets the non-retentive TON.

Engineering Purpose: The timer creates the implemented delay between acceptance of a run request and physical conveyor startup.

8.11 Rung 0010 - Conveyor Motor Output

Ladder Logic:

```
|----[ XIC T4:0/DN ]------[ XIC B3:0/2 ]-----[ XIO B3:0/6 ]----------( OTE O:2/0 )----|
      Startup_Delay_DN     Conveyor_Run       Batch_Complete         Conveyor_Motor
```

Rung Description: Conveyor_Motor O:2/0 energizes only after the startup timer is complete, Conveyor_Run remains true, and Batch_Complete is false. Loss of any condition de-energizes the output.

Engineering Purpose: Rechecking Conveyor_Run prevents operation from a stale timer state, while Batch_Complete directly stops production at the target quantity.

8.12 Rung 0011 - Running Lamp

Ladder Logic:

```
|----[ XIC O:2/0 ]-------------------------------------( OTE O:2/1 )----|
     Conveyor_Motor                                     Running_Lamp
```

Rung Description: Running_Lamp O:2/1 follows the Conveyor_Motor command.

Engineering Purpose: I used the commanded motor state for a simple operator indication of conveyor operation.

8.13 Rung 0012 - Load Batch Target

Ladder Logic:

```
|-----------------------------------------------------[ MOV ]----|
                                                       Source: N7:0
                                                       Dest:   C5:0.PRE
```

Rung Description: The unconditional MOV continuously copies N7:0 into C5:0.PRE. The current screenshot shows C5:0.PRE = 3 because N7:0 is set to 3 during the captured test state.

Engineering Purpose: I used an integer target register so the required batch quantity can be changed without editing the CTU rung and can later be supplied by an HMI.

8.14 Rung 0013 - Photoeye Batch Counter

Ladder Logic:

```
|----[ XIC O:2/0 ]----[ XIC I:1/7 ]----[ ONS B3:2/0 ]----[ CTU C5:0 ]----|
     Conveyor_Motor     Photoeye          One_Shot        Batch_Counter
```

Rung Description: Products are counted only while Conveyor_Motor is ON. A false-to-true Photoeye transition passes through ONS B3:2/0 and increments C5:0 once.

Engineering Purpose: Motor qualification prevents stopped-conveyor counts, and the one-shot prevents one sustained Photoeye signal from being counted on multiple PLC scans.

8.15 Rung 0014 - Batch Complete Latch

Ladder Logic:

```
|----[ XIC C5:0/DN ]----------------------------------( OTL B3:0/6 )----|
     Batch_Counter_Done                                 Batch_Complete
```

Rung Description: When C5:0.ACC reaches C5:0.PRE, C5:0/DN becomes true and Batch_Complete B3:0/6 latches.

Engineering Purpose: I latched the end-of-batch state so completion remains visible and continues to inhibit operation until the deliberate reset sequence is performed.

8.16 Rung 0015 - Batch Complete Lamp

Ladder Logic:

```
|----[ XIC B3:0/6 ]-----------------------------------( OTE O:2/4 )----|
     Batch_Complete                                   Batch_Complete_Lamp
```

Rung Description: Batch_Complete_Lamp O:2/4 energizes whenever the latched Batch_Complete bit is true.

Engineering Purpose: This rung provides a dedicated operator indication that the configured production quantity has been reached.

8.17 Rung 0016 - Batch Counter Reset

Ladder Logic:

```
|----[ XIC I:1/2 ]----[ XIC B3:0/6 ]-----------------[ RES C5:0 ]----|
       Reset_PB           Batch_Complete                Batch_Counter
```

Rung Description: When Reset_PB is pressed while Batch_Complete is true, RES C5:0 clears the counter accumulator and counter status bits.

Engineering Purpose: Conditioning the reset with Batch_Complete prevents this normal reset path from clearing the count during an active batch.

8.18 Rung 0017 - Batch Complete Reset

Ladder Logic:

```
|----[ XIC I:1/2 ]----[ XIC B3:0/6 ]-----------------( OTU B3:0/6 )----|
      Reset_PB           Batch_Complete                   Batch_Complete
```

Rung Description: The same reset action unlatches Batch_Complete. Because Rung 0016 is scanned first, the counter reset is evaluated before Rung 0017 removes Batch_Complete.

Engineering Purpose: I used separate RES and OTU rungs so the batch recovery sequence and scan order are easy to understand and troubleshoot.

8.19 Rung 0018 - Jam Detection Timer

Ladder Logic:

```
|----[ XIC O:2/0 ]----[ XIO I:1/7 ]-------------------[ TON T4:1 ]----|
      Conveyor_Motor     Photoeye                     Jam_Timer  Time Base: 1.0  PRE: 10
```

Rung Description: While Conveyor_Motor is ON and Photoeye remains false, Jam_Timer T4:1 accumulates. A Photoeye detection makes the XIO instruction false and resets the TON. If no detection occurs for ten seconds, T4:1/DN becomes true.

Engineering Purpose: I used this timer as a simple no-product-flow diagnostic. The preset represents the expected maximum interval without a product detection for this simulation.

8.20 Rung 0019 - Jam Fault Latch

Ladder Logic:

```
|----[ XIC T4:1/DN ]---------------------------------( OTL B3:1/0 )----|
     Jam_Timer_Done                                      Jam_Fault
```

Rung Description: When T4:1/DN becomes true, Jam_Fault B3:1/0 latches. Jam_Fault then becomes one of the master fault sources in Rung 0001.

Engineering Purpose: I latched the jam diagnostic so the event is retained for deliberate recovery rather than disappearing immediately when the conveyor stops.

8.21 Rung 0020 - Alarm Horn

Ladder Logic:

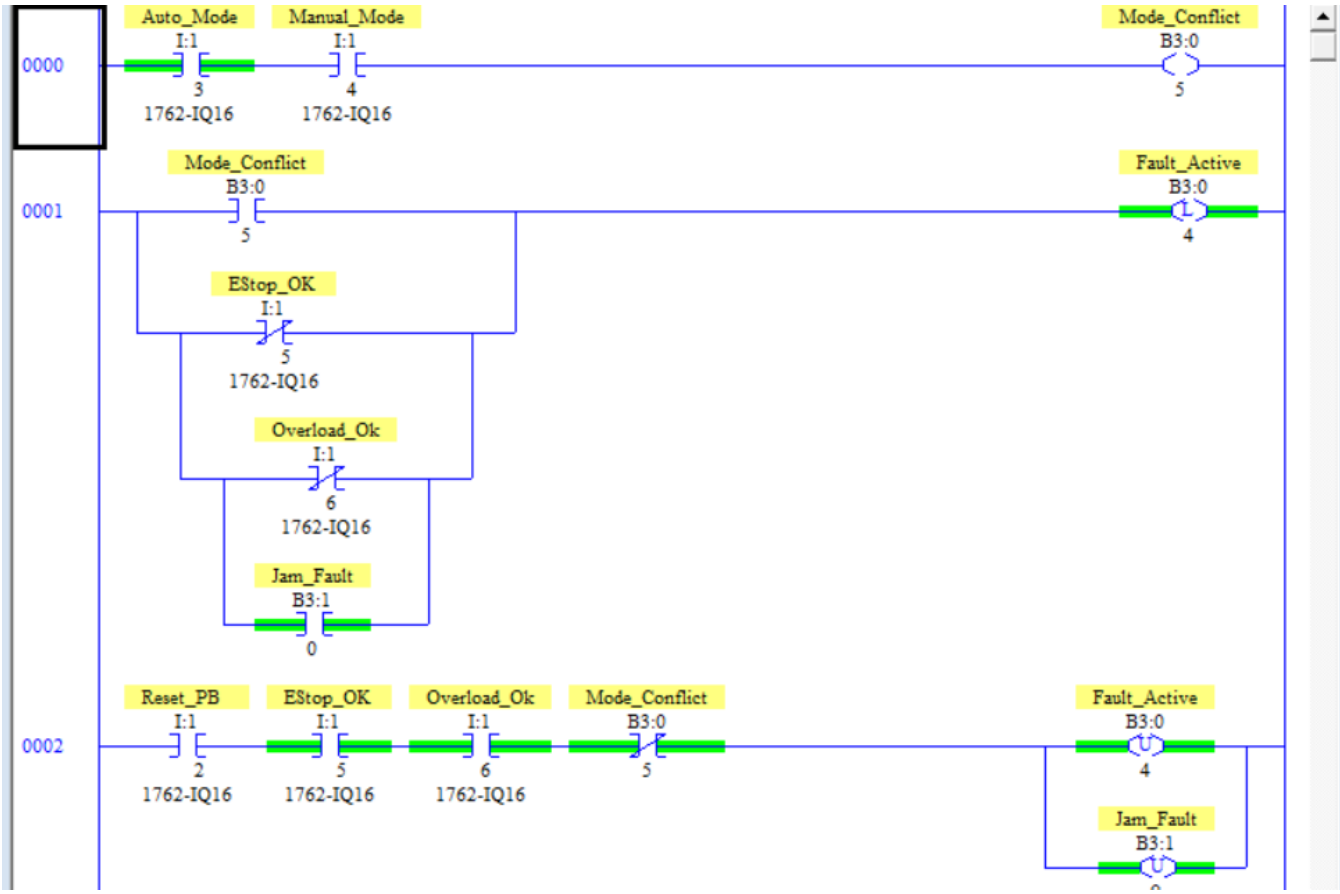
```
|----[ XIC B3:0/4 ]----------------------------------( OTE O:2/3 )----|
     Fault_Active                                    Alarm_Horn
```

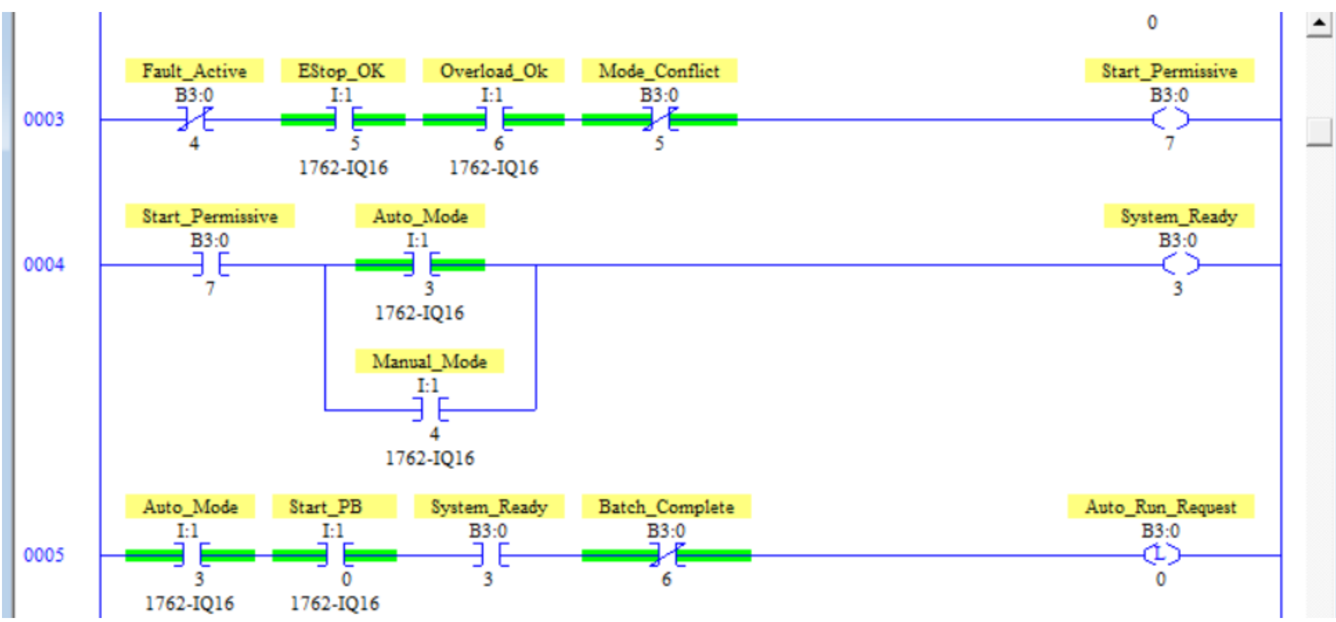
Rung Description: Alarm_Horn O:2/3 follows the latched Fault_Active state and remains energized until the master fault is successfully reset.

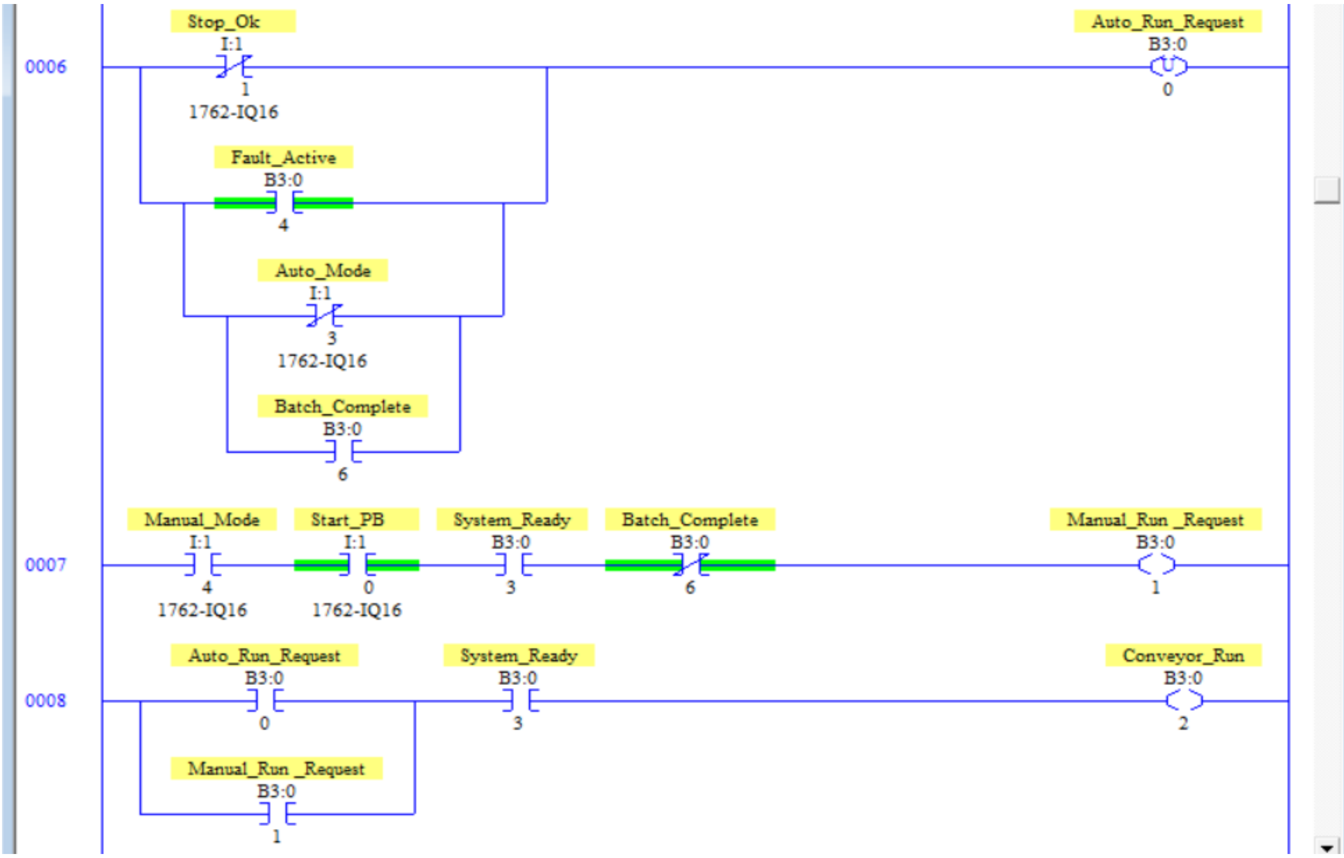
Engineering Purpose: I used the master fault as the audible alarm source so Mode Conflict, E-stop, overload, and jam events produce one consistent alarm response.

8.22 Actual RSLogix Implementation

The following figures show the final RSLogix program used as the technical source of truth for this report. The screenshots are included as implementation evidence and correspond to the typed rung descriptions above.






0006
Stop_Ok
I:1
1
1762-IQ16
Fault_Active
B3:0
4
Auto_Mode
I:1
3
1762-IQ16
Batch_Complete
B3:0
6
Auto_Run_Request
B3:0
U
0
0007
Manual_Mode
I:1
4
1762-IQ16
Start_PB
I:1
0
1762-IQ16
System_Ready
B3:0
3
Batch_Complete
B3:0
6
Manual_Run _Request
B3:0
1
0008
Auto_Run_Request
B3:0
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Manual_Run _Request
B3:0
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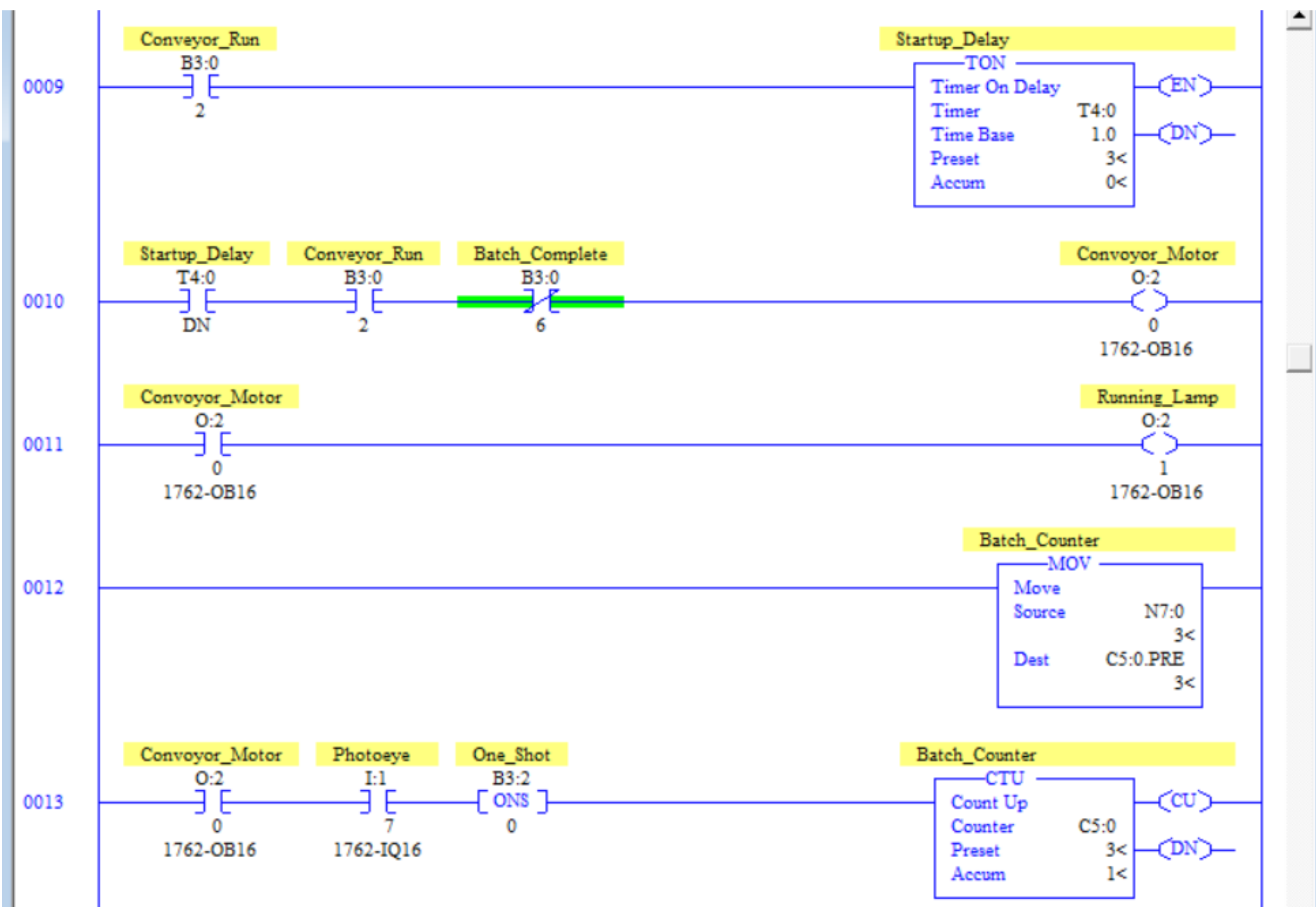
System_Ready
B3:0
3
Conveyor_Run
B3:0
2


0009
Conveyor_Run
B3:0
2
Startup_Delay
TON
Timer On Delay
Timer T4:0
Time Base 1.0
Preset 3<
Accum 0<
EN
DN
0010
Startup_Delay
T4:0
DN
Conveyor_Run
B3:0
2
Batch_Complete
B3:0
6
Convoyor_Motor
O:2
0
1762-OB16
0011
Convoyor_Motor
O:2
0
1762-OB16
Running_Lamp
O:2
1
1762-OB16
0012
Batch_Counter
MOV
Move
Source N7:0
3<
Dest C5:0.PRE
3<
0013
Convoyor_Motor
O:2
0
1762-OB16
Photoeye
I:1
7
1762-IQ16
One_Shot
B3:2
ONS
0
Batch_Counter
CTU
Count Up
Counter C5:0
Preset 3<
Accum 1<
CU
DN

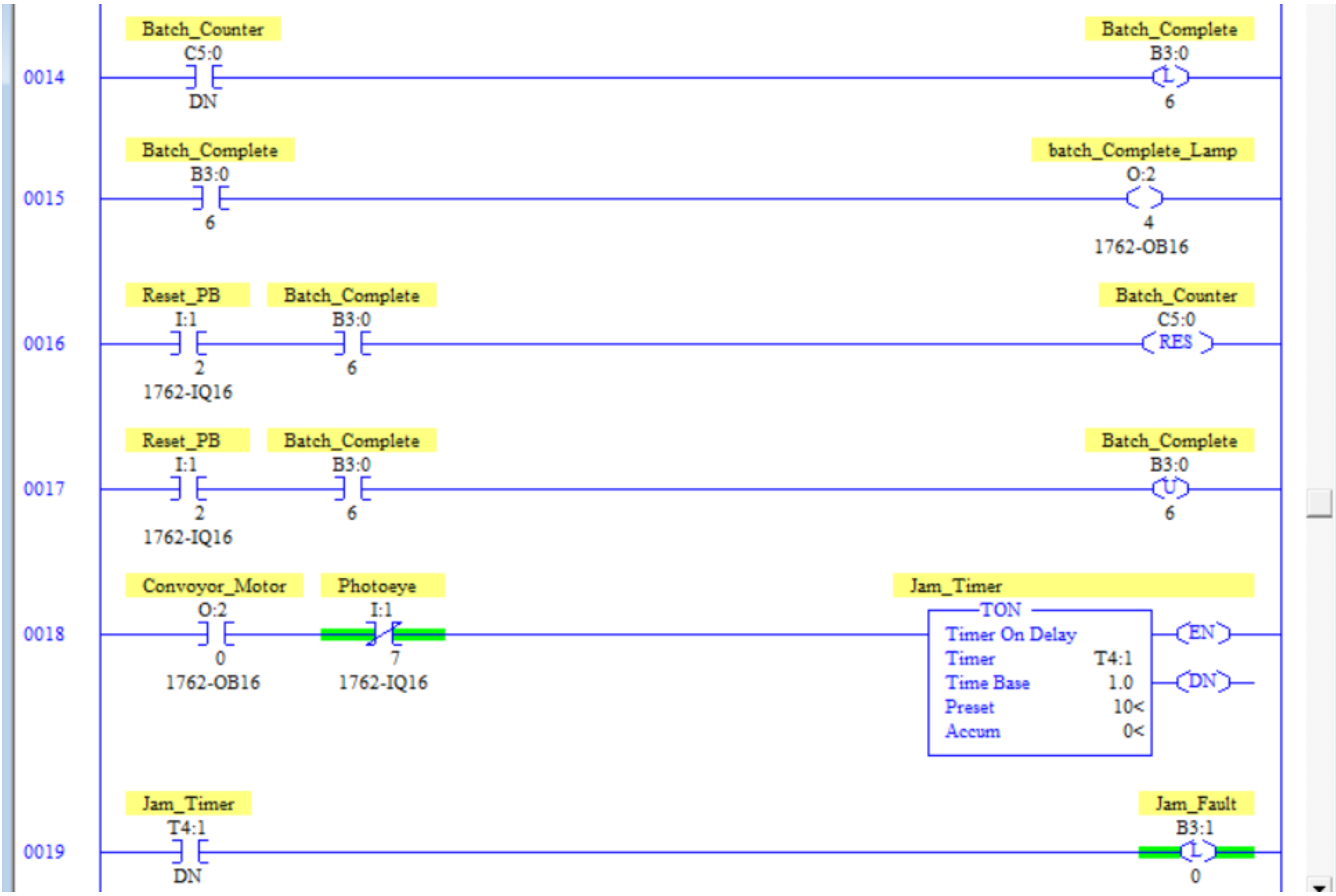


Figure P2V2.5 - Final RSLogix Implementation, Part 5

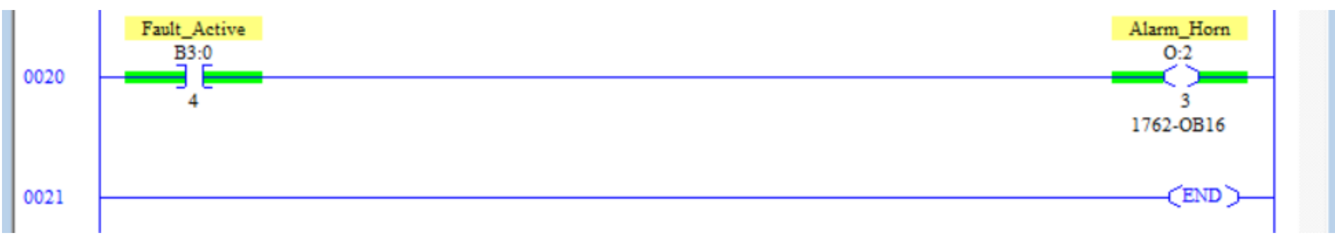


Figure P2V2.6 - Final RSLogix Implementation, Part 6

9. Sequence of Operation

Step 1 - Initial Condition: The PLC evaluates the current fault state, E-stop and overload healthy signals, and operating-mode selection. Conveyor_Motor remains OFF until the startup sequence is satisfied.

Step 2 - Mode Selection: The operator selects either Auto mode or Manual mode.

Step 3 - Mode Conflict Check: If Auto and Manual are selected simultaneously, Mode_Conflict B3:0/5 becomes true.

Step 4 - Master Fault Check: Mode Conflict, an unhealthy E-stop input, an unhealthy overload input, or Jam_Fault B3:1/0 can latch Fault_Active B3:0/4.

Step 5 - Start Permissive: When Fault_Active is false, EStop_OK and Overload_OK are healthy, and no Mode_Conflict is present, Start_Permissive B3:0/7 becomes true.

Step 6 - System Ready: With Start_Permissive true and either Auto or Manual selected, System_Ready B3:0/3 becomes true.

Step 7 - Auto Run Request: In Auto mode, Start_PB latches Auto_Run_Request when System_Ready is true and Batch_Complete is false.

Step 8 - Manual Run Request: In Manual mode, Start_PB energizes the non-latched Manual_Run_Request while System_Ready is true and Batch_Complete is false.

Step 9 - Common Conveyor Run: Either mode-specific request can establish Conveyor_Run while System_Ready remains true.

Step 10 - Startup Delay: Conveyor_Run enables T4:0 and the timer accumulates for three seconds.

Step 11 - Conveyor Start: When T4:0/DN is true, Conveyor_Run remains true, and Batch_Complete is false, Conveyor_Motor O:2/0 energizes.

Step 12 - Running Indication: Running_Lamp O:2/1 follows Conveyor_Motor.

Step 13 - Batch Target Loading: The unconditional MOV continuously transfers N7:0 into C5:0.PRE.

Step 14 - Product Detection: While Conveyor_Motor is ON, each valid Photoeye transition is processed through ONS B3:2/0.

Step 15 - Product Counting: The one-shot pulse increments Batch_Counter C5:0 once per product.

Step 16 - Batch Completion: When C5:0.ACC reaches C5:0.PRE, C5:0/DN becomes true and latches Batch_Complete B3:0/6.

Step 17 - Conveyor Stop at Batch Completion: Batch_Complete inhibits Conveyor_Motor and also participates in the Auto/Manual request logic.

Step 18 - Batch Complete Indication: Batch_Complete_Lamp O:2/4 energizes while Batch_Complete is true.

Step 19 - Batch Counter Reset: With Batch_Complete true, pressing Reset executes RES C5:0.

Step 20 - Batch Complete Reset: The following rung unlatches Batch_Complete so a new batch can begin.

Step 21 - Jam Monitoring: While Conveyor_Motor is ON and Photoeye remains false, Jam_Timer T4:1 accumulates.

Step 22 - Jam Fault: If Photoeye remains false for the full ten-second preset, T4:1/DN latches Jam_Fault B3:1/0.

Step 23 - Master Fault Response: Jam_Fault causes Fault_Active to latch; Start_Permissive drops and the conveyor stops through the request/run logic.

Step 24 - Alarm Indication: Alarm_Horn O:2/3 energizes whenever Fault_Active is true.

Step 25 - Fault Recovery: After the fault-producing condition is corrected, Reset can unlatch Fault_Active and Jam_Fault when the Rung 0002 conditions are satisfied.

10. Testing & Validation

I tested Project 2 Version 2 in the PLC development/simulation environment. The supplied project documentation records successful completion of normal startup, batch counting, batch reset, jam detection, jam recovery, and safety-interlock verification. The final ladder screenshot is the tested source of truth, so the verification table below is aligned to that implementation.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Select one valid mode with healthy E-stop/overload conditions	Start_Permissive and System_Ready become true; motor remains OFF	Ready-state logic operated as required	PASS
T-02	Select Auto and Manual simultaneously	Mode_Conflict becomes true and Fault_Active latches	Mode conflict correctly detected	PASS
T-03	In Auto mode, press Start	Auto_Run_Request latches	Automatic request remained after Start release	PASS
T-04	Maintain valid Auto request	T4:0 accumulates for 3 s before conveyor starts	Startup delay completed correctly	PASS
T-05	Allow T4:0/DN to become true	Conveyor_Motor and Running_Lamp energize	Conveyor started after delay	PASS
T-06	Generate one Photoeye transition while motor is ON	C5:0 increments once	Product count increased by one	PASS
T-07	Hold Photoeye true across multiple scans	ONS prevents repeated count	Single detection produced one count	PASS
T-08	Count until C5:0 reaches C5:0.PRE	Batch_Complete latches	Batch complete activated at target	PASS
T-09	Observe conveyor after Batch_Complete	Conveyor_Motor turns OFF	Conveyor stopped automatically	PASS
T-10	Observe Batch_Complete_Lamp	O:2/4 turns ON	Completion indication operated	PASS
T-11	Press Reset after completed batch	C5:0 resets and Batch_Complete clears	Batch reset completed successfully	PASS
T-12	Run conveyor with Photoeye remaining false	T4:1 accumulates toward 10 s	Jam timer accumulated correctly	PASS
T-13	Keep Photoeye false until T4:1/DN	Jam_Fault and Fault_Active latch	Jam condition correctly detected	PASS
T-14	Correct jam condition and press Reset	Fault_Active and Jam_Fault clear	Jam recovery completed successfully	PASS
T-15	Make EStop_OK false	Fault_Active latches and operation is inhibited	E-stop monitoring response verified	PASS
T-16	Make Overload_OK false	Fault_Active latches and operation is inhibited	Overload response verified	PASS
T-17	Make Stop_OK false during Auto operation	Auto_Run_Request unlatches and conveyor stops	Auto stop behavior verified	PASS

T-18	Remove Auto mode while Auto request is active	Auto_Run_Request unlatches	Mode-removal shutdown verified	PASS
T-19	Operate in Manual mode	Manual_Run_Request follows the non-latched command path	Manual operation verified	PASS
T-20	Change N7:0	MOV updates C5:0.PRE to the target value	Dynamic target function verified	PASS

Validation Summary

Testing confirmed that the final program performed the intended conveyor-control, batch-counting, jam-monitoring, fault-handling, and reset functions.

The conveyor did not start immediately after a valid request; the configured three-second delay completed first. Product counts occurred only while the conveyor motor was commanded ON, and the one-shot prevented a sustained Photoeye signal from generating repeated counts.

The batch target was supplied through N7:0 and moved into C5:0.PRE. Reaching the preset latched Batch_Complete, stopped the conveyor, and energized the Batch Complete Lamp. The reset sequence cleared both the counter and completion latch.

Jam detection operated by timing continuous conveyor operation without a Photoeye detection. When T4:1 reached its ten-second preset, Jam_Fault and then Fault_Active latched. After the condition was corrected, the implemented reset sequence returned the system to normal operation.

This testing represents functional PLC program validation in the project development/simulation environment, not production-machine commissioning or safety-system validation.

11. Results

Project 2 Version 2 successfully expanded the original batch-counting concept into a structured conveyor-control application.

The final program supports Auto and Manual modes, Mode Conflict detection, centralized Start_Permissive and System_Ready states, and separation between operating requests and the physical Conveyor_Motor output.

Automatic operation uses a maintained Auto_Run_Request, while Manual operation uses a non-latched request. Both modes share Conveyor_Run and the three-second startup delay.

The production-counting logic uses Photoeye only while the conveyor motor is energized. ONS B3:2/0 ensures one counter increment per valid product-detection transition. N7:0 supplies the dynamic batch target through the MOV to C5:0.PRE.

When the configured quantity is reached, Batch_Complete latches, stops the conveyor, and energizes Batch_Complete_Lamp O:2/4. The operator can then reset the counter and completion state for the next batch.

The jam-monitoring section adds a process diagnostic that was not part of the foundation design. Continuous conveyor operation without a Photoeye detection for ten seconds latches Jam_Fault and activates the master fault architecture.

All defined functional requirements passed in the final PLC testing environment.

12. Challenges & Troubleshooting

A useful troubleshooting path for this project is: Mode / Protective Inputs -> Mode_Conflict / Fault_Active -> Start_Permissive -> System_Ready -> Auto/Manual Request -> Conveyor_Run -> T4:0 -> Conveyor_Motor -> Photoeye -> C5:0 -> Batch_Complete.

For jam-related problems, the diagnostic path continues: Conveyor_Motor -> Photoeye -> T4:1 -> Jam_Fault -> Fault_Active -> Alarm_Horn.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
System Ready does not turn ON	Fault active, unhealthy E-stop/overload input, or no mode selected	Monitor B3:0/4, I:1/5, I:1/6, I:1/3, I:1/4	Restore required conditions and select a valid mode
Auto request will not latch	Auto mode, Start, System Ready, or Batch Complete condition incorrect	Trace Rung 0005	Verify each enabling contact
Auto request drops unexpectedly	Stop_OK false, Fault Active, Auto mode removed, or Batch Complete active	Monitor all Rung 0006 branches	Identify the shutdown branch that became true
Manual request disappears	Start or System Ready removed, or Batch Complete active	Monitor Rung 0007	Verify Manual mode and all enabling conditions
Startup timer does not accumulate	Conveyor_Run is false	Monitor B3:0/2 and T4:0/EN	Trace request and System Ready logic
Timer done but conveyor OFF	Conveyor_Run false or Batch Complete true	Monitor T4:0/DN, B3:0/2, B3:0/6	Restore run request or complete batch reset
Counter does not increment	Motor OFF, Photoeye does not transition, or one-shot not firing	Monitor O:2/0, I:1/7, B3:2/0, C5:0.ACC	Verify conveyor state and sensor transition
Counter increments only once in repeated test	Photoeye never returned false between products	Monitor I:1/7	Ensure a complete OFF-to-ON transition for each product
Batch completes at wrong quantity	Unexpected N7:0 value	Compare N7:0 and C5:0.PRE	Correct Target Quantity
Batch Complete Lamp does not turn ON	Batch_Complete not set or output rung false	Monitor B3:0/6 and O:2/4	Verify C5:0/DN and Rung 0015
Counter does not reset	Reset or Batch_Complete condition missing	Trace Rung 0016	Verify I:1/2 and B3:0/6
Jam fault occurs	No product detection for 10 s while conveyor runs	Monitor O:2/0, I:1/7, T4:1	Verify product flow and Photoeye operation
Jam fault returns after reset	No-product condition still exists	Monitor Photoeye and T4:1	Correct process/sensor condition before reset
Alarm Horn remains ON	Fault_Active remains latched	Monitor B3:0/4	Identify fault source and perform valid reset

One of the main troubleshooting lessons from this project was that a counting problem is not always a counter problem. The full path includes the conveyor state, Photoeye transition, one-shot behavior, counter preset, Batch_Complete state, and the target value stored in N7:0.

13. Lessons Learned

This project strengthened my understanding of how product counting fits into a complete conveyor-control sequence rather than existing as an independent CTU rung.

I learned how Auto and Manual operating requests can be separated from the physical output and then combined into one common Conveyor_Run signal. This reduced duplicated downstream control logic and made the startup sequence easier to follow.

The project reinforced the purpose of centralized permissive and readiness logic. Start_Permissive and System_Ready provide defined points for checking whether the implemented operating conditions are satisfied before a request reaches the motor logic.

Using N7:0 as the batch target demonstrated how an integer register can separate a production setpoint from the ladder instruction itself and create a natural future HMI interface.

The Photoeye counting logic reinforced the importance of one-shot instructions. A product can remain in front of a sensor for many PLC scans, so the count should be associated with the event transition rather than the duration of the signal.

I also learned how Batch_Complete can be used as an operating state rather than only an indication. It influences the request and motor logic and remains latched until the deliberate batch-reset sequence is performed.

Finally, jam detection demonstrated how timer-based diagnostics can identify abnormal process behavior by comparing expected product flow with actual sensor activity.

14. Industrial Relevance

Conveyor-based batch counting is used in manufacturing, packaging, warehouse automation, food processing, pharmaceutical production, parcel handling, assembly operations, and other material-handling systems.

This Version 2 project demonstrates controls concepts relevant to those applications: mode management, permissive logic, maintained and momentary operating requests, delayed startup, configurable production targets, sensor-based counting, batch completion, automatic production stopping, jam monitoring, fault handling, and operator indication.

The configurable target architecture is particularly relevant because production quantities are commonly supplied from an HMI, recipe system, production order, or higher-level control system rather than being permanently coded into a counter instruction.

The project remains a PLC training and portfolio implementation. A real production conveyor would require additional engineering appropriate to the application, including safety-rated stopping functions, motor protection and feedback, guarding, sensor validation, upstream/downstream coordination, and commissioning on the actual equipment.

15. Version 1 -> Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Benefit
Basic conveyor start/stop	Auto and Manual operating modes	Adds defined operating behaviors
Direct operating logic	Separate mode requests and common Conveyor_Run	Separates operator intention from physical output
Limited interlocks	Start_Permissive and System_Ready	Improves control structure and diagnostics
Fixed batch target	N7:0 -> C5:0.PRE	Allows configurable production quantity
Product counting	Motor-qualified Photoeye + ONS + CTU	Improves count validity
Batch indication only	Latched Batch_Complete integrated into control	Creates a defined end-of-batch state
Conveyor could continue after target	Batch_Complete inhibits motor and Auto request	Stops production at target quantity
Simple reset	Controlled counter and Batch_Complete reset	Prevents normal active-batch reset
No jam detection	T4:1 no-product timer	Adds process-flow diagnostics
No persistent jam fault	Latched Jam_Fault	Preserves abnormal condition for recovery
Limited fault handling	Latched Fault_Active	Provides common abnormal-state handling
Limited status indication	Running Lamp, Batch Complete Lamp, Alarm Horn	Improves operator visibility

The main progression is not simply additional rungs. Version 2 changes the project from a basic conveyor counter into a sequence that manages operating requests, validates readiness, delays startup, uses a configurable target, stops at batch completion, monitors product flow, and provides defined fault/recovery behavior.

16. Project Summary

Project 2 Version 2 upgraded the original conveyor-counting project into an Industrial Batch Conveyor Counter System using an Allen-Bradley MicroLogix 1100 PLC and RSLogix Micro Starter Lite.

The final ladder includes Auto and Manual modes, Mode Conflict detection, Start_Permissive, System_Ready, separate operating requests, a common Conveyor_Run signal, a three-second startup delay, conveyor output control, product counting, a configurable target, Batch_Complete handling, jam detection, fault latching, reset logic, and operator indication.

Target Quantity N7:0 is continuously transferred to C5:0.PRE. Products are counted only while Conveyor_Motor is energized, and ONS B3:2/0 ensures each valid Photoeye transition increments C5:0 once.

When C5:0 reaches the configured target, Batch_Complete latches, Conveyor_Motor stops, and Batch_Complete_Lamp energizes. Reset then clears the counter and completion latch to prepare for the next batch.

Jam_Timer T4:1 detects extended conveyor operation without a Photoeye detection. At the ten-second timeout, Jam_Fault latches and activates the master Fault_Active architecture and Alarm_Horn.

All defined requirements passed in the PLC project/simulation environment. The completed Version 2 project demonstrates my progression from simple conveyor counting to a more structured control system using operating modes, permissives, production states, configurable quantities, diagnostics, faults, and controlled recovery.